

Marko Stefanov

EDUCATION

University College London

Bachelor of Computer Science

London, UK

Sep. 2024 – Exp. Jun. 2027

- Current Mark: 81.02/100.00
- Fields of study: Systems and Software Engineering, Mathematics for Computer Science, Programming, Security, Computer Architecture and Concurrency, Intelligent Systems (Minor).

Mathematics-Informatics High-School

Secondary School

Skopje, MK

Sep. 2020 – Jun. 2024

- Average: 5.0/5.0

TECHNICAL SKILLS

Programming Languages: C++, C, Python, Java, Haskell, HTML, CSS, Prolog, Clojure

Tools and Frameworks: Django, Arduino Boards, Espressif Boards, Bazel

Languages: English (fluent), Macedonian (native), Serbo-Croatian (intermediate)

PROJECTS

ReadingStar 2.0 – Releasing Soon

- Collaborated with **IBM**, **Intel** and the **NAS**.
- Developed a karaoke-style application to help students with autism gain confidence in speaking aloud.
- Created the music generation leveraging Meta's **MusicGen - Small** text-to-music model.

PLC for Biogas Power Plant

- Developed control and safety systems for a reserve generator used during main generator maintenance.
- Programmed a **Programmable Logic Controller** with an integrated **Human-Machine Interface** to enable safe and user-friendly generator operation.
- Designed and implemented software and hardware interlocks to prevent simultaneous use by multiple processes, ensuring equipment protection and compliance with safety standards.

Tuberculosis Vaccine Production Bioreactor Model

- Led a team of eight students to design and build a fully functional small-scale bioreactor.
- Delivered a control system enabling real-time management of temperature, pH, and stirring speed via a custom application layer interface.
- Implemented a **MQTT** publisher-subscriber communication between the frontend and an **ESP32** microcontroller, with **I²C** bus to an **Arduino** for actuator control (heating element, stirring motor, alkali and acid pumps).

Profil – Web App for Industrial Use

- Developed a web app to generate optimal pipe-cutting instructions using the column generation algorithm.
- Application was **successfully integrated** at ElektroLab, minimising industrial waste.

Minesweeper Bot

- Created a bot to solve Google Minesweeper using **OpenCV** and **PyAutoGUI**.
- Designed an algorithm that completes over **99.99%** of Minesweeper levels with definite solutions.

Tetris Autoplayer AI

- Designed a Tetris AI using ten heuristics and **reinforcement learning** to achieve competitive scores.

UCL Yellow Pages Web App

- Developed a website to assist UCL students in finding their peers' contact information.
- Implemented the platform using the **Django** framework in **Python**, with data managed through **SQL (Django ORM)**.

EXPERIENCE

Arm – GPU Software Intern

Jun. 2026 – Sept. 2026

- Owned three public-source image building **Bazel rulesets** for building embedded Linux artefacts, including GPT disk images, filesystem images and U-Boot images, with **reproducible toolchain support** and **tests**.
- Implemented a development to release pipeline including **GitLab CI**, **pre-commit** checks, **release setup**, **licensing** and **end-to-end tests**.

SmartFinance – Intern

Jul. 2025 – Aug. 2025

- Reviewed business plans and assisted clients in securing government funding.
- Gained exposure to **M&A** processes, **financial documentation**, and **accounting**.

Man-Dra – Intern

Jul. 2023 – Aug. 2023

- Coded **Programmable Logic Controllers** to read and display data from electrical measuring instruments using ladder logic.
- Developed a **Human-Machine Interface** to allow users to interact with the Programmable Logic Controller intuitively.

ElektroLab – Intern

Jul. 2020

- Constructed a power distribution panel from concept to completion.
- Programmed electrical panels using analog and digital inputs/outputs.